

Bulevirtide monotherapy in patients with chronic HDV needs further evaluation

To the Editor,

We were particularly intrigued by the recent article by Wedemeyer *et al.* in the *Journal*.¹ This article not only represents a significant academic contribution but also guides the medical profession. The 96-week clinical trial assessing the efficacy and safety of bulevirtide (BLV) monotherapy for patients with chronic hepatitis D (CHD) revealed consistent improvements in virological and biochemical responses. Importantly, BLV treatment maintained a consistent safety profile throughout the 96 weeks, with no new safety concerns. While the article provides valuable insights and showcases the author's expertise, we observed certain areas that may warrant attention and refinement.

First, while the article primarily evaluates the efficacy of BLV monotherapy in patients with CHD, addressing the urgent need for effective treatment in this rare and severe condition, it lacks histological data. Liver biopsy is crucial for definitively diagnosing cirrhosis and providing detailed insights into liver pathology.² For instance, Allweiss *et al.* found through a combined analysis of paired liver biopsies from three clinical trials that BLV treatment can decrease the number of HDV-infected hepatocytes and reduce markers of liver inflammation.³ The absence of biopsy data may hinder a comprehensive understanding of patients' liver conditions, potentially affecting the accurate assessment of treatment outcomes. Furthermore, magnetic resonance elastography and other non-invasive tests (NITs) for evaluating hepatic fibrosis could complement biopsy findings and enhance the accuracy of liver lesion assessment. For example, studies by Sandman *et al.* have demonstrated the effectiveness of NITs in diagnosing cirrhosis and clinically significant portal hypertension in HDV.^{4,5} Future studies should focus on enhancing data collection and analysis to provide a thorough understanding of treatment effects on liver health.

Second, response to BLV treatment can vary based on individual factors such as genetic background and immune status. Genetic background may impact drug metabolism and action, while immune status can influence the immune response to treatment. It is important to note that BLV targets the NTCP receptor on hepatocytes rather than a viral component. Previous studies have shown that viral resistance is unlikely to occur with BLV treatment, as it inhibits the virus's entry into hepatocytes by blocking the NTCP receptor.⁶ This mechanism reduces the likelihood of viral resistance, making BLV a promising option for long-term therapy. Further studies are needed to explore how individual factors influence the efficacy of BLV treatment and improve personalized treatments and outcomes.

Third, interferons are pivotal in treating CHD, primarily through their immunomodulatory and antiviral actions. Immunomodulation entails enhancing natural killer cell activity,

promoting T-cell activation, and bolstering the antiviral immune response, thereby strengthening the host's resistance against HDV and minimizing liver damage. The antiviral effect directly impedes HDV replication, inhibiting its reproduction within the liver and aiding in disease progression control.⁷ Recent studies by Asselah *et al.* have performed direct comparisons between BLV monotherapy and interferon treatment, providing valuable insights into their respective advantages and limitations.⁸ These analyses may offer deeper insights into the benefits and drawbacks of BLV in CHD treatment, thereby informing clinical practice and treatment strategies.

Certain data in the article were obtained through patient self-reports. For treatment adherence, patient diaries were utilized to document medication usage details, encompassing the dispensed number of study medication vials and the returned number of empty vials. This reliance on patient self-reporting introduces subjectivity and uncertainty, potentially affecting data accuracy and reliability. The MEMS (Medication Event Monitoring System), a cutting-edge tool for evaluating patient adherence to treatment, incorporates an electronic device into the bottle cap and records the precise timing of each medication bottle opening. These data provide researchers with intricate insights into the medication behavior of patients, facilitating a comprehensive understanding of treatment adherence patterns.⁹ Additionally, measuring drug plasma/urine concentration provides an objective and accurate method to assess if the medication was taken as prescribed, surpassing the reliability of self-reporting or questionnaires.¹⁰ By incorporating objective adherence assessment methods, combining multiple techniques, and ensuring data validation and verification could effectively address the limitations of the article, thereby improving data reliability and accuracy.

In conclusion, despite these limitations that merit careful consideration, the study provides a useful starting point for scholarly discussion and suggests directions and issues for future research, which may even be conducted applying the data and material obtained within the MYR-301 and/or other registration trials. We would like to express our sincere gratitude to Wedemeyer *et al.* for their invaluable contributions to this area of research.

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Conflict of interest

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Authors' contributions

JL penned the initial draft of the manuscript, which was subsequently scrutinized by GX. YT then conducted thorough revisions on the paper. The manuscript's final version has been reviewed and authorized for publication by all authors.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.07.023>.

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